Unit 4 2019.pdf

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5. *(a)* State the laws of Faraday and Lenz for electromagnetic induction. [2]

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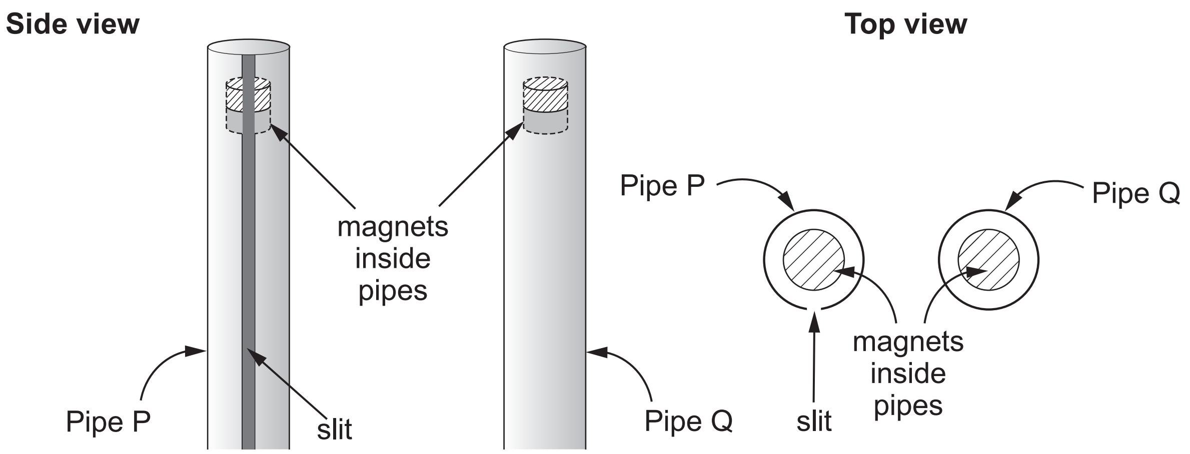
(b) Two strong bar magnets are dropped through two copper pipes (P and Q). Pipe P has a slit running along its length but Q is complete. When the magnet is dropped through pipe P it accelerates almost uniformly but the magnet dropped through pipe Q quickly reaches a very low terminal velocity. Explain these observations. [6 QER]



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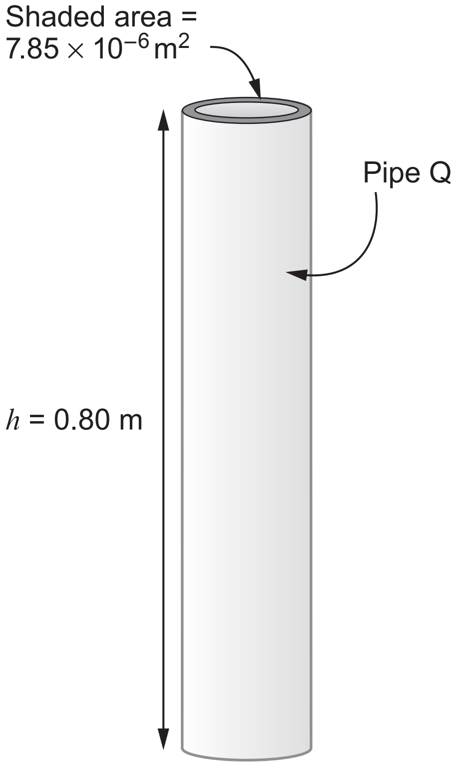
(c) By applying the principle of conservation of energy, calculate the temperature increase of pipe Q after the magnet has fallen at constant speed. [4]

Data:

- mass of magnet = 0.300 kg,
- cross-sectional area of the copper walls of pipe Q = $7.85 \times 10^{-6}\,m^2$ (see diagram)
- density of copper = $8\,960\,kg\,m^{-3}$
- specific heat capacity of copper = $385\,J\,K^{-1}\,kg^{-1}$.



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